

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

CY1003-1 – MATERIALS CHEMISTRY FOR COMPUTER SYSTEMS

(For AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Note:
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided.
Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The device which converts electrical energy into chemical energy
A) Galvanic Cell
B) Fuel cell
C) Electrolytic Cell
D) Solar cell
2. What is the electrode potential of the calomel electrode with saturated KCl solution at 298K?
A) 0.334 V
B) 0.281 V
C) 0.242 V
D) 0.0591 V
3. Which of the following is NOT a component of battery?
A) Cathode
B) Salt-bridge
C) Anode
D) Electrolyte
4. Special properties of 'Li' metal that make it advantageous as an electrode material include
A) Light weight
B) Low electrical conductivity
C) High electrode potential
D) Low cost
5. Which of the following is suitable for powering portable devices such as smartphones and laptops?
A) Lead-acid battery
B) Lithium-ion battery
C) Methanol-oxygen fuel cell
D) Vanadium flow battery
6. The total number of functional groups or bonding sites present in a monomer molecule is called
A) Functionality
B) Degree of Polymerization
C) Molecular Weight
D) Polymer
7. Which of the following is a monomer?
A) Ethane
B) Ethyl chloride
C) Ethene
D) Ethyl alcohol
8. In the conductometric titration of, the conductance increases gradually till equivalence point and then rises sharply.
A) Strong acid vs strong base
B) Strong acid vs weak base
C) Weak acid vs strong base
D) Weak acid vs weak base
9. The driving force in electroless plating isreaction on a pretreated active surface.
A) Autocatalytic Oxidation
B) Autocatalytic reduction
C) Autocatalytic displacement
D) Autocatalytic redox
10. The reference electrode used in the electrochemical gas sensing of NO_x is
A) Ag/AgCl
B) Zn/ZnCl₂
C) Fe/FeCl₂
D) Cu/CuSO₄
11. 1 nm =
A) 10^{-7} m
B) 10^{-9} m
C) 10^9 m
D) 10^{-3} m

12. Which is the preferred precursor in the sol-gel method?
 A) Metal chlorides B) Metal nitrates
 C) Metal hydroxides D) Metal alkoxides
13. ROM is defined as
 A) Read Out Memory B) Read Once Memory
 C) Read Only Memory D) Read One Memory
14. Which computer memory chip allows simultaneous both read and write operations?
 A) ROM B) RAM
 C) PROM D) EEPROM
15. Which property of polymers makes them suitable for applications in memory devices?
 A) High electrical resistance B) Low thermal stability
 C) Bistable state D) Brittle nature
16. Liquid crystal displays (LCDs) work based on the principle of:
 A) Electromagnetic induction B) Electrochemical reactions
 C) Opto-electronic effect D) Liquid crystal polarization
17. Recycling of e-waste has a direct impact on
 A) Environment B) Economy
 C) Both environment and economy D) Neither environment nor economy
18. Extraction of metals by treatment with HCl is called as leaching.
 A) Cyanide B) Thiourea
 C) Acid D) Thiosulphate
19. In which process are the electronic components harvested from e-waste without breaking them into smaller components?
 A) Hydrometallurgy B) Direct recycling
 C) Pyrometallurgy D) Electrolysis
20. Splitting of water into hydrogen and oxygen using energy is called
 A) Fission B) Fusion
 C) Electrolysis D) Photosynthesis

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define reference electrode. Explain the construction and working of calomel electrode.	8	L*2	1	1
	b) Explain the construction and working of lithium-ion battery. Mention its applications.	8	L2	1	1
2.	a) Explain the mechanism of conduction in polyacetylene.	7	L2	2	1
	b) Calculate the number and weight average molecular weights of a polymer with 200 molecules of molecular mass 500; 300 molecules of molecular mass 2000 and 500 molecules of molecular mass 3500.	6	L3	2	2
	c) Mention the advantages of instrumental techniques over traditional methods.	3	L1	2	1
3.	a) How is glass electrode used to determine the pH of unknown solution?	4	L2	1	1
	b) Calculate the cell potential of the following cell at 298K $\text{Ag} \text{Ag}^+(0.001\text{M}) \text{Ag}^+(0.50\text{M}) \text{Ag}$	3	L3	1	2
	c) Describe the basic principle and working of conductometry for the estimation of strong and weak acid.	9	L2	2	1
Unit – II					
4.	a) Highlight the principle governing electrochemical sensors. How is it used to detect SO _x ?	6	L2	3	1
	b) Demonstrate the process of PCB fabrication using electroless plating technique. Mention the advantages of electroless plating.	10	L3	3	1

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| 5. a) | Describe the electronic memory storing capacity of organic molecules by taking the example of pentacene. | 8 | L2 | 4 | 1 |
| b) | Explain the opto-electronic effect observed in LCs. Demonstrate their application in LCDs. | 8 | L3 | 4 | 1 |
| 6. a) | Discuss the working of amperometric and optical sensors in the determination of DO. | 6 | L2 | 3 | 1 |
| b) | Classify the electronic memory based on volatility. | 10 | L1 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | Explain the steps involved in the hydrometallurgical extraction method for metal recovery from e-waste. | 8 | L2 | 5 | 1 |
| b) | Enumerate the properties of hydrogen fuel. Demonstrate the generation of green hydrogen by PEM method. | 8 | L1 | 5 | 1 |
| 8. a) | Explain the construction and working of solar photovoltaic (PV) cell. What are the advantages of PV cell? | 9 | L2 | 5 | 1 |
| b) | Explain the stages involved in the process of gold recovery from e-waste. | 7 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
